

APEX RETAIL GROUP

FROM STORE RANKINGS TO ENTERPRISE INSIGHT

Professional Portfolio Project Report

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Business context	60 stores • 5 regions • 3 brands • 24 months
Core model	Stores • Categories • Sales • Targets • DateTable
Advanced features	Time intelligence • RANKX • ALLEXCEPT • Field Parameters • What-If • Drill-through • Tooltips • Bookmarks • RLS
Decision focus	Store performance • profitability • target attainment • Enterprise Zone impact

1. Project Overview

I built this Power BI project to move a retail performance conversation beyond simple store rankings and towards enterprise insight. The analytical model covers 60 stores, five regions, three brands, 24 months and eight product categories. The final solution connects commercial performance, profitability, target attainment, regional peer comparison and Enterprise Zone adjustments in a single decision-support experience.

BUSINESS QUESTION	Which stores and categories require closer management attention, and how does the answer change when performance is viewed relative to regional peers and approved Enterprise Zone relief?
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2. Solution Architecture

I structured the solution as a star-schema Power BI model with Stores and Categories as dimensions, Sales and Targets as fact tables, and DateTable as the shared time dimension. I standardised source data types in Power Query and created two calculated store attributes: Store Age and Size Band.

Model component	Role
Stores	Store, brand, region, type, size and Enterprise Zone attributes
Categories	Category names, margin benchmarks and brand affinity
Sales	24-month store-category sales, cost, discount and gross profit fact
Targets	Monthly revenue targets by store
DateTable	Continuous date dimension supporting time intelligence

3. Analytical Layer and DAX

I built a reusable Measures table rather than embedding calculations inside individual visuals. The foundation layer covers revenue, cost, gross profit, gross margin, discount, units, transactions, basket size, revenue per square foot and target variance. I then extended the layer with time intelligence, dynamic rankings and Enterprise Zone-adjusted profitability.

Time intelligence

- YTD revenue and gross profit
- SPLY revenue and margin
- Revenue vs SPLY %
- Prior month and MoM % change
- Rolling three-month revenue analysis
- Cumulative revenue

Dynamic ranking

I used RANKX to support both company-wide and regional views. The company rank uses ALL(Stores); the regional rank uses ALLEXCEPT(Stores, Stores[region]) so each store is compared with its regional peers rather than the full company. I also created regional average revenue, variance-to-region-average and Bottom Quartile logic.

4. Interactive Reporting

I designed the report as five connected analytical pages with a common navigation rail and consistent Apex Noir styling. The interaction model combines field parameters, What-If analysis, drill-through, report-page tooltips and bookmarks so that users can move from enterprise context to detailed investigation without leaving the report.

Executive Summary

Headline KPIs, 24-month revenue/SPLY trend, regional comparison and dynamic performance metric selection.

Store Performance & Rankings

Store-level company and regional ranking, Bottom Quartile classification, brand/region slicers and drill-through entry point.

Margin & Category Analysis

Rolling three-month category margin, category benchmark comparison, What-If margin simulation and Revenue vs Target variance by store.

Enterprise Zone Impact

Before/After bookmark comparison, adjusted ranking, rank change and relief impact.

Store Profile

24-month revenue/profit trend, category revenue, target variance and store metadata.

5. Enterprise Zone Analysis

I modelled the Enterprise Zone scenario as an explicit cost restatement rather than a manual visual adjustment. Six stores carry an 8% source-cost loading and receive approved relief percentages of 14-22%. The EZ Adjusted Cost measure uses SUMX and RELATED, with adjusted gross profit and gross margin measures recalculated from the restated cost.

VALIDATED IMPACT	Non-EZ cost remained £144.69M. EZ cost reduced from £17.51M to £14.40M, a reduction of approximately £3.11M across the dataset.
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I then recalculated regional margin ranking and Bottom Quartile status using the adjusted profitability measure. The before-and-after view makes the effect of the business rule visible at store level and supports a more controlled interpretation of store performance.

6. Security and Controlled Access

I implemented six RLS roles: five regional manager roles and an unrestricted Commercial Director role. Each regional role filters Stores[region] to the relevant region. I tested the model through View As and retained role configuration evidence.

7. Visual Design and User Experience

I used an Apex Noir design system built around deep charcoal backgrounds, dark cards, champagne-gold accents and restrained status colours. The left navigation rail creates a consistent user journey, while page headers, insight annotations and compact KPI cards establish a clear visual hierarchy. On dense analytical pages I deliberately reduced decorative elements so the analytical content remains primary.

DESIGN PRINCIPLE	The dashboard is designed to answer the next business question quickly: what changed, where is the exception, why does it matter, and where should the user investigate next?
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8. Validation and Evidence

I validated the analytical model against the supplied source data and tested the main interactive features in Power BI Desktop. The South East revenue cross-check matched £69,000,812.21. The model also reconciled to approximately £247.75M total revenue and 34.5% gross margin at the report level.

Capability	Implemented result
Time intelligence	24-month trend, SPLY, YTD, rolling, MoM and cumulative measures
Dynamic ranking	Company and regional RANKX with Bottom Quartile logic
Field parameter	Controlled switching across four performance measures
What-If analysis	0-10% margin improvement parameter in 0.5-point increments
Drill-through	Store Profile reached from store ranking context
Tooltip	Store ranking hover context with trend and regional rank
Bookmarks	Before/After Enterprise Zone profitability and ranking views
RLS	Five regional roles plus Commercial Director

9. Business Insight Summary

Commercial performance

The dashboard headline is £247.75M revenue, with Revenue vs SPLY at -2.5% and gross margin at 34.5%. The analysis therefore moves beyond topline scale and focuses on relative store performance, profitability and target attainment.

Category and margin management

Category profitability is visible through benchmark comparison and rolling three-month margin. The What-If layer allows a controlled sensitivity view of margin improvement without changing the underlying actuals.

Enterprise Zone interpretation

The Enterprise Zone adjustment materially changes the effective cost base for the six affected stores and can change their regional profitability ranking. Making the adjustment explicit prevents the business rule from being hidden in manual interpretation.

10. Project Outcome

I completed the solution as an end-to-end Power BI analytical product: from source preparation and star-schema modelling through reusable DAX, validation, interactive reporting, Enterprise Zone scenario analysis and RLS. The final dashboard provides a structured path from enterprise-level context to store-level investigation and supports different stakeholder questions without duplicating the underlying analytical logic.

PORTFOLIO SUMMARY	A decision-focused Power BI solution combining modelling, DAX, advanced interactivity, security and business storytelling in one coherent retail analytics product.
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Appendix - Core DAX Catalogue

Measure	Measure	Measure
Total Revenue	Total Cost	Gross Profit
Gross Margin %	Total Discount	Discount Rate %
Total Units	Total Transactions	Avg Basket Size
Revenue per Sq Ft	Revenue Target	Variance to Target
Variance to Target %	Revenue YTD	Revenue SPLY
Revenue vs SPLY %	Gross Profit YTD	Gross Margin % SPLY
Revenue Rolling 3M	Revenue Prior Month	Revenue MoM % Change
Revenue Cumulative	Store Revenue Rank (Company)	Store Margin Rank (Company)
Store Revenue Rank (Region)	Store Margin Rank (Region)	Region Avg Revenue
vs Region Avg %	Is Bottom Quartile (Region)	Dynamic Chart Title
Store EZ Status	EZ Tax Relief %	Adjusted Gross Margin %
EZ Adjusted Cost	EZ Adjusted Gross Profit	EZ Adjusted Gross Margin %
EZ Relief Impact (ppts)	Store Margin Rank - EZ Adjusted (Region)	Is Bottom Quartile - EZ Adjusted
Rank Change - EZ Adjustment		